

ELARIS-ONE

Campus Intelligence Platform

Phase 1 — Design & Functional MVP Progress

B.Tech CSE (AI & ML) • Project Progress Presentation

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1. Problem Statement & Project Objective

Why Elaris-One is being developed

Problem

College resources are often scattered across WhatsApp groups, drives, chats and separate portals. Students spend time searching for notes, PYQs, opportunities and academic guidance.

Objective

Build one campus-focused platform that centralizes academic resources and opportunities, while adding an AI layer to help students discover and understand information faster

2. Proposed Solution

Elaris-One acts as a single digital workspace for campus intelligence

Centralized academic resources — Notes and Previous Year Questions (PYQs)

Opportunity hub — internships, jobs, hackathons and events

Personal productivity — bookmarks for saved resources

Student identity — profile and academic information

AI assistance — conversational interface for campus-related help

Protected user experience — authentication and route protection

3. Technology Stack & Architecture

Technologies used in the current implementation

Frontend

React + TypeScript
Vite
React Router
Axios

Backend

Node.js
Express
TypeScript
REST APIs

Database

PostgreSQL
Prisma ORM

AI Layer

Gemini service integration
AI Assistant / Chat UI

Authentication → protected routes → API services → backend controllers/services → database

Modular frontend structure: pages, components, context, routes, services and types

Git + GitHub used for version control and project collaboration

4. Phase 1 Progress & Current Status

What has been completed so far

Completed

- ✓ Project structure
- ✓ Frontend UI foundation
- ✓ Authentication screens
- ✓ Dashboard
- ✓ Notes module
- ✓ PYQ module
- ✓ Opportunities module
- ✓ Bookmarks module
- ✓ Profile page
- ✓ Protected routing
- ✓ API/service layer

Working Integration

- ✓ Frontend ↔ backend API communication
- ✓ Notes/PYQs/Opportunities data retrieval
- ✓ Bookmark add/remove flow
- ✓ PDF/resource links
- ✓ User profile display
- ✓ AI service/backend foundation
- ✓ Production frontend build verified

Phase 1 Status

DESIGN + FUNCTIONAL MVP FOUNDATION

The platform is no longer only a UI prototype. Core pages are connected to backend services and key user flows are demonstrable.

Next phase: deeper AI intelligence, richer data, admin workflows and deployment.

5. Features Working Right Now

Current demo-ready functionality

Login / Registration — user authentication flow

Dashboard — overview of available resources and quick navigation

Notes — resource listing, search/filtering, subject information and PDF opening

PYQs — previous-year question listing with semester/branch/year information

Opportunities — internships, jobs, hackathons and events with application links

Bookmarks — save resources, view saved items and remove bookmarks

Profile — student identity, role and academic information display

AI Assistant / AI Chat — current AI interface and backend integration foundation

6. Future Enhancements — Phase 2 & Beyond

Features planned after the design/MVP phase

AI Intelligence

- Campus-aware RAG over notes/PYQs
- AI resource discovery
- Smart study plans
- Question answering from uploaded material
- Personalized recommendations

Platform Expansion

- Admin/faculty dashboard
- Resource upload & moderation
- Better search and filters
- Richer student profiles

7. Conclusion & Next Roadmap

Phase 1 establishes the foundation for Elaris-One

Current milestone: a functional, navigable campus platform with core modules implemented.

The UI/design system is established across the major student-facing pages.

Backend APIs, authentication, bookmarks and resource flows provide the working foundation.

AI integration has been started and will become the major intelligence layer in the next phase.

Immediate next step: strengthen AI/RAG, admin workflows, data quality, testing and deployment.

Long-term goal: make Elaris-One the intelligent operating layer for a college campus.

Demo flow: Login → Dashboard → Notes → PYQs → Opportunities → Bookmark → Profile → AI Assistant